# ANDROID PROGRAMMING PRACTICAL 06

## Module 1 · Communication between Activity and Fragment

### Aim

Demonstrate communication between an Activity and a Fragment.

**Project name**: ActivityFragmentCommunicationDemo

### Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

### Procedure

**1.** Create MainActivity and a MessageFragment with a TextView and an input/button.

**2.** Add the fragment to a FragmentContainerView in the activity layout.

**3.** Define a callback interface in the Fragment, or use a shared ViewModel for communication.

**4.** Have the Fragment send a user-entered message through the callback when its button is pressed.

**5.** Implement the callback in MainActivity and update an Activity TextView or show a Toast.

**6.** If using a shared ViewModel, observe LiveData/StateFlow from both components and update the value from the Fragment.

**7.** Run the app and verify messages pass from Fragment to Activity; test empty input handling.

### Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

### Expected result

A message or value is passed between the Fragment and Activity and reflected in the UI.

### Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

### Conclusion

The stated Android concept was implemented and verified through the steps and tests above.